Big Data Analytics Project

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1. INTRODUCTION

Credit risk assessment is one of the most important analytical tasks in financial services because it helps lenders estimate the likelihood that a borrower will fail to meet repayment obligations. In practice, this affects loan approval, pricing, portfolio quality, and overall risk management. In my Milestone 1 proposal, I defined the project around classification and predictive analytics using the Home Credit Default Risk dataset, with the aim of identifying the drivers of default and evaluating models that can support data-informed lending decisions.

This second milestone extends that proposal by reviewing relevant prior research, describing the selected dataset in more detail, and presenting a structured methodology for the project. The capstone course expects this milestone to include a literature review, preliminary dataset analysis, and a tentative research approach, so this document is designed to directly support those requirements.

2. LITERATURE REVIEW

2.1 TRADITIONAL CREDIT SCORING AND CLASSIFICATION

Credit scoring and default prediction have long been treated as supervised classification problems. Traditional methods such as logistic regression remain important because they are relatively interpretable and have strong acceptance in financial risk practice. At the same time, more advanced machine learning methods such as decision trees, random forests, gradient boosting, and neural-network-based models are increasingly studied because they can capture nonlinear relationships and interactions among borrower features. The Home Credit competition itself frames the task as predicting how capable each applicant is of repaying a loan, which makes it a direct fit for classification-based credit risk analysis.

2.2 MACHINE LEARNING IN CREDIT RISK

A recurring finding in credit-risk research is that machine learning models can outperform simpler baseline models in predictive accuracy, especially when datasets contain complex interactions, many variables, and nonlinear relationships. For example, supervisory research published through the European Banking Authority compared several machine learning approaches for credit default prediction and found that more flexible models could improve predictive performance relative to simpler traditional benchmarks, although interpretability and governance remained important concerns. This is relevant to the current project because the Home Credit dataset is rich, multi-table, and likely to benefit from models that can capture complex borrower patterns.

2.3 CLASS IMBALANCE

Another major issue in credit-risk modelling is **class imbalance**. In many lending datasets, default cases are fewer than non-default cases, which can cause models to favor the majority class and miss risky borrowers. Chawla et al.'s SMOTE paper is

foundational in this area and showed that synthetic oversampling of the minority class, especially when combined with majority-class undersampling, can improve classifier performance in ROC space. More recent credit-risk studies continue to emphasize that imbalance handling is important when the goal is not only overall accuracy but also stronger identification of risky borrowers. This matters for my project because a lender is usually more concerned with correctly identifying likely default cases than with maximizing raw accuracy alone.

2.4 INTERPRETABILITY AND EXPLAINABILITY

Interpretability is another major theme in the literature. In credit decision-making, a high-performing model is not enough on its own; users also need to understand which variables drive risk predictions. Lundberg and Lee introduced SHAP as a unified feature-attribution framework for explaining model predictions, and later work in credit scoring has highlighted the value of explainable AI for balancing performance with transparency. This is especially important in financial settings, where decision-makers need to justify model outputs, understand variable importance, and communicate findings clearly to stakeholders. For this reason, the current project will not stop at model training; it will also include feature importance analysis and interpretable reporting.

2.5 LITERATURE GAP AND PROJECT POSITIONING

The literature also suggests that credit-risk projects are strongest when they combine three elements: careful preprocessing, robust model comparison, and practical business reporting. Many published studies compare baseline interpretable models with stronger nonlinear models, evaluate them using more than one metric, and then translate the findings into decision-support insights. This directly supports the structure of my capstone project: first understand the data, then compare models, then present findings through dashboards and visualizations that are meaningful in a financial context. Although the literature on credit scoring is extensive, there is still value in applying these ideas to the Home Credit dataset in a structured capstone setting. First, many studies emphasize predictive performance but spend less time showing a clean end-to-end workflow from dataset understanding to stakeholder-oriented reporting. Second, some research focuses heavily on advanced models without clearly comparing them to interpretable baselines. Third, explainability is often discussed conceptually but not always integrated into the practical reporting layer. My project addresses these gaps by combining:

1. detailed dataset handling,
2. baseline and machine learning model comparison, and
3. interpretable reporting through feature importance and dashboard-style outputs. This direction is also consistent with the methodology and reporting expectations of the course.

3. Summary of Prior Research

Study	Method	Key Finding	Relevance to This Project
Chawla et al. (2002)	SMOTE	Improves classification on imbalanced data	Helps handle default imbalance
Lundberg & Lee (2017)	SHAP	Explains model predictions	Used for interpretability
EBA (2020)	ML comparison	ML > traditional models	Supports model selection
Noriega et al. (2023)	Review study	Feature engineering is critical	Guides preprocessing

The reviewed studies are relevant because they address problems similar to this project, including credit risk prediction, imbalanced classification, model interpretability, and the importance of preprocessing. These findings help justify the methods selected in this capstone.

4. Dataset Description

The selected dataset is the **Home Credit Default Risk** dataset from Kaggle, which was also identified in Milestone 1 as the core dataset for the project. It is a structured consumer lending dataset designed for a supervised learning task in which the objective is to predict whether an applicant will have repayment difficulties. The main application data are split into training and test files, where the training file includes the target variable and the test file does not. Kaggle also provides linked tables containing historical and behavioral information, such as previous applications, bureau records, installment payments, POS cash balance, and credit card balance. (A preview of the main application training dataset is shown in Figure 1)

```
import pandas as pd

df = pd.read_csv('application_train.csv')
df.head()
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	A
0	100002	1	Cash loans	M	N	Y	0	202500.0	406597.5	
1	100003	0	Cash loans	F	N	N	0	270000.0	1293502.5	
2	100004	0	Revolving loans	M	Y	Y	0	67500.0	135000.0	
3	100006	0	Cash loans	F	N	Y	0	135000.0	312682.5	
4	100007	0	Cash loans	M	N	Y	0	121500.0	513000.0	

5 rows × 122 columns

Figure 1. Preview of the Home Credit application training dataset

As shown in Figure 1, the dataset includes a large number of variables and a mix of demographic, financial, and application-related attributes. The presence of the TARGET field confirms that the dataset is appropriate for supervised classification.

A separate dataset description file was used to understand the meaning and business context of each variable. This helped ensure accurate interpretation of features during data analysis and preprocessing.

For example:

- `AMT_INCOME_TOTAL` represents the applicant's total income
- `AMT_CREDIT` represents the loan amount
- `CNT_CHILDREN` indicates the number of children of the applicant

Understanding the business meaning of variables allows for better interpretation of patterns and supports more meaningful feature engineering in later stages.

4.1 Dataset Source

The dataset used in this project is the Home Credit Default Risk dataset, publicly available on Kaggle. It can be accessed at:
https://www.kaggle.com/competitions/home-credit-default-risk/data

This dataset is suitable for the capstone for several reasons:

- First, it directly matches the project's classification theme because the target variable represents whether the applicant experienced repayment difficulty.
- Second, it is large and realistic enough to support meaningful predictive analysis.
- Third, it contains a mix of demographic, financial, and historical behavioral variables, which makes it useful for feature analysis as well as model development.
- Finally, the dataset is publicly available and widely used, which makes it appropriate for a course-based applied analytics project.

4.2Main Data Characteristics

The dataset contains both **numerical** and **categorical** variables. (The size of the main training dataset is shown in Figure 2.)

```
df.shape

(307511, 122)
```

```
Show code

DATASET OVERVIEW
----------------------------------------
Rows        : 307,511
Columns     : 122
Numerical   : 106
Categorical: 16
```

Figure 2. Shape of the application training dataset

Figure 2 shows the number of rows and columns in the main training file. The relatively high number of features increases the complexity of preprocessing and supports the need for structured feature selection and cleaning.
Numerical fields include values such as income, credit amount, annuity, and age-related or employment-related time measures. Categorical fields include items such as contract type, education level, housing type, and occupation-related attributes. In addition, the linked tables provide temporal and behavioral history that can be aggregated into useful borrower-level features. As a result, the dataset is not only appropriate for straightforward classification, but also for feature engineering and cross-table aggregation.

4.3 Constraints and Limitations

The dataset also has important limitations. It represents a specific lending environment and therefore may not generalize directly to all banking systems or regulatory contexts. This limitation was already acknowledged in Milestone 1. In addition, the dataset is complex and includes multiple related tables, so improper joins or weak preprocessing could reduce model quality. Another likely challenge is missing data, because real-world financial datasets often contain incomplete fields due to unavailable customer information, documentation gaps, or historical system differences. Finally, because the target is a default-related outcome, class imbalance is expected to be a methodological concern that should be addressed during modeling and evaluation rather than ignored.

4.4 Planned Data Handling and Cleaning

To meet the Milestone 2 rubric, the dataset handling process must be documented clearly and justified. The planned data handling steps are as follows:

1. **Data loading and structure review**
   The first step will be to load the main application training file and inspect dimensions, column names, data types, and target distribution. Related tables will then be reviewed to understand how they connect to the main applicant file. This supports a clean analytical foundation before modeling begins.

```
df.describe()
```

	SK_ID_CURR	TARGET	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	REGION_POPULATI
count	307511.000000	307511.000000	307511.000000	3.075110e+05	3.075110e+05	307499.000000	3.072330e+05	3
mean	278180.518577	0.080729	0.417052	1.687979e+05	5.990260e+05	27108.573909	5.383962e+05	
std	102790.175348	0.272419	0.722121	2.371231e+05	4.024908e+05	14493.737315	3.694465e+05	
min	100002.000000	0.000000	0.000000	2.565000e+04	4.500000e+04	1615.500000	4.050000e+04	
25%	189145.500000	0.000000	0.000000	1.125000e+05	2.700000e+05	16524.000000	2.385000e+05	
50%	278202.000000	0.000000	0.000000	1.471500e+05	5.135310e+05	24903.000000	4.500000e+05	
75%	367142.500000	0.000000	1.000000	2.025000e+05	8.086500e+05	34596.000000	6.795000e+05	
max	456255.000000	1.000000	19.000000	1.170000e+08	4.050000e+06	258025.500000	4.050000e+06	

8 rows × 106 columns

2. **Missing value analysis**
   Missingness will be examined by variable and by percentage.(An initial overview of missing values is shown in Figure 4.)

```
missing = df.isnull().sum().sort_values(ascending=False)
missing.head(10)
```

	0
COMMONAREA_AVG	214865
COMMONAREA_MODE	214865
COMMONAREA_MEDI	214865
NONLIVINGAPARTMENTS_MEDI	213514
NONLIVINGAPARTMENTS_MODE	213514
NONLIVINGAPARTMENTS_AVG	213514
FONDKAPREMONT_MODE	210295
LIVINGAPARTMENTS_AVG	210199
LIVINGAPARTMENTS_MEDI	210199
LIVINGAPARTMENTS_MODE	210199

dtype: int64

Figure 4. Variables with the highest missing-value counts

Figure 4 indicates that missing data is a meaningful issue in this dataset and confirms the need for careful imputation or removal strategies depending on the importance and missingness level of each variable.

3. Variables with extremely high missingness may be removed if they contribute little analytical value, while variables with moderate missingness may be imputed. Numerical variables may be imputed using median values, while categorical variables may use mode or an explicit “Unknown” category, depending on the variable meaning.

3. Abnormal and outlier handling

Outliers will be identified through summary statistics and visual inspection. However, outlier treatment in credit data must be done carefully. Extreme values may represent genuine high-risk borrowers rather than data errors, so the project will distinguish between data-entry anomalies and legitimate but unusual financial cases. If abnormal

values appear impossible or clearly inconsistent, they may be capped, transformed, or excluded with justification.

4. Categorical encoding
Categorical variables will be transformed into model-ready form. Low-cardinality categories may be handled using one-hot encoding, while higher-cardinality variables may require more selective handling to avoid unnecessary sparsity.

5. Feature engineering and aggregation
Where appropriate, related tables will be aggregated to the applicant level using counts, averages, ratios, delays, or trend-based summaries. This is especially important because repayment behavior is often better captured through historical patterns than through static application variables alone.

6. Train-validation split and reproducibility
The training data will be split into training and validation subsets using a fixed random state so that model comparison remains reproducible and fair.

7. Class imbalance treatment
If the target variable is notably imbalanced, the project will test strategies such as class weights or SMOTE on the training set only. This is supported by prior research showing the importance of imbalance-aware learning for credit-risk problems.

This cleaning and handling plan directly addresses the rubric requirement to describe dataset characteristics, constraints, cleaning steps, and the handling of abnormalities with justification.

4.5 Summary Statistics

Preliminary analysis of the dataset includes summary statistics of key numerical variables such as income, credit amount, and annuity. These statistics provide insight into the scale, spread, and potential skewness of important financial fields. As shown in Figure 3, the numerical variables vary substantially in magnitude, indicating that applicants differ widely in their financial profiles. The summary statistics also suggest the presence of extreme values, which will need to be reviewed carefully to distinguish valid high-value cases from possible anomalies.

```
df.describe()
```

	SK_ID_CURR	TARGET	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	REGIC
count	307511.000000	307511.000000	307511.000000	3.075110e+05	3.075110e+05	307499.000000	3.072330e+05	
mean	278180.518577	0.080729	0.417052	1.687979e+05	5.990260e+05	27108.573909	5.383962e+05	
std	102790.175348	0.272419	0.722121	2.371231e+05	4.024908e+05	14493.737315	3.694465e+05	
min	100002.000000	0.000000	0.000000	2.565000e+04	4.500000e+04	1615.500000	4.050000e+04	
25%	189145.500000	0.000000	0.000000	1.125000e+05	2.700000e+05	16524.000000	2.385000e+05	
50%	278202.000000	0.000000	0.000000	1.471500e+05	5.135310e+05	24903.000000	4.500000e+05	
75%	367142.500000	0.000000	1.000000	2.025000e+05	8.086500e+05	34596.000000	6.795000e+05	
max	456255.000000	1.000000	19.000000	1.170000e+08	4.050000e+06	258025.500000	4.050000e+06	

8 rows × 106 columns

Figure 3. Summary statistics for selected numerical variables

As shown in Figure 3, key numerical variables such as applicant income, credit amount, and annuity display substantial variation in scale and range. This suggests that the dataset contains financially diverse applicants and may also include extreme values that should be reviewed during preprocessing. The summary statistics also indicate that several variables are likely to be skewed rather than normally distributed.

4.6 Correlation Analysis

Correlation analysis will be conducted to examine relationships between numerical variables and their association with the target variable (default risk). The strongest correlations with the target variable among numerical features are shown in Figure 7.

Figure 7. Top numerical correlations with the target variable

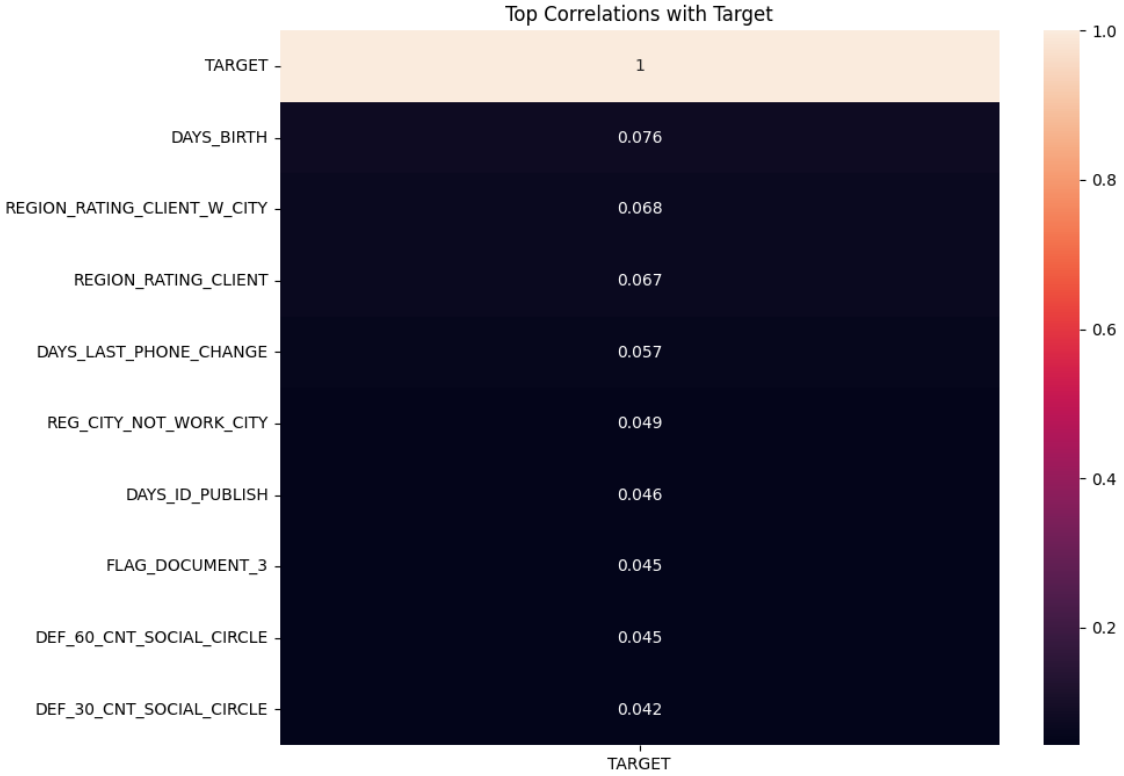

Figure 7. Top numerical correlations with the target variable

As shown in Figure 7, the numerical features with the strongest relationship to the target variable are not necessarily the same as the features with the strongest correlations among themselves. The observed correlations with TARGET appear relatively modest, which is common in credit-risk datasets where default behaviour is influenced by combinations of variables rather than a single dominant feature. This supports the use of multivariate modelling approaches rather than relying on individual feature relationships alone.In particular, the variables appearing at the top of Figure 7 will be investigated further in later modelling stages.

4.7 Abnormal Data Identification

Abnormal or inconsistent data points are identified using statistical methods and domain understanding.

These values are handled carefully to distinguish between:

- True high-risk cases

- Data quality issues

Abnormal or inconsistent data points were identified conceptually through initial data exploration and will be further examined using summary statistics, missing-value analysis, and visual inspection. Potential examples include unusually high income values, unrealistic employment-related durations, and incomplete categorical records. These observations do not automatically imply errors; some may represent genuine high-risk or unusual borrower situations. Therefore, abnormal values will be evaluated carefully before deciding whether to retain, transform, cap, or exclude them.

4.8 Key Observations from Data

Initial analysis of the dataset reveals several important patterns (The raw class counts for the target variable are shown in Figure 5.)

```
df['TARGET'].value_counts()
```

TARGET	count
0	282686
1	24825

dtype: int64

Figure 5. Count of default and non-default cases in the target variable

The class imbalance is also illustrated visually in Figure 6.

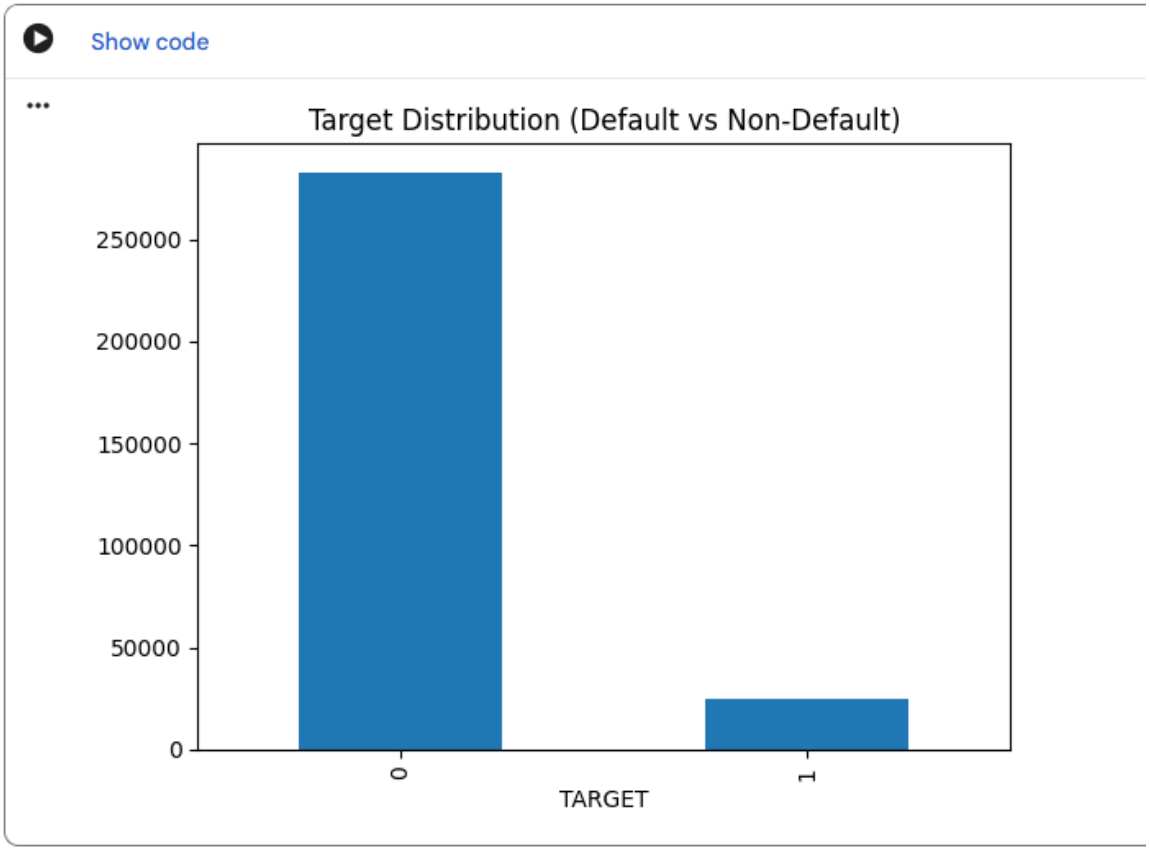

Figure 6. Distribution of the target variable

Figures 5 and 6 confirm that non-default cases substantially outnumber default cases. This imbalance is important because it affects model evaluation and justifies the later use of metrics beyond accuracy.

Initial exploration of the dataset reveals several important patterns. First, the target variable is imbalanced, with non-default cases significantly outnumbering default cases, as shown in Figures 5 and 6. Second, financial variables such as income, credit amount, and annuity appear important, but no single variable is expected to fully explain default behaviour on its own. Third, missing values are common across several attributes, reinforcing the need for careful preprocessing. Finally, the availability of linked historical tables suggests that behavioural and repayment-history features may improve predictive performance in later stages of the project.

5. RESEARCH QUESTIONS AND JUSTIFICATION

The project continues with the three interconnected research questions defined in Milestone 1:

1. **Which borrower and loan-related features are most associated with higher default risk?**
2. **Can classification models accurately predict the likelihood of loan default?**
3. **How can credit risk predictions be presented through dashboards and reporting to support financial decision-making?**

These questions remain relevant and feasible for Milestone 2 because they connect the project's analytical, predictive, and practical goals.

The first question is important because identifying the main drivers of risk helps explain the problem before modeling. It also supports later interpretation through feature importance, which makes the results more useful for a non-technical audience.

The second question addresses model performance directly. Because credit default prediction is a classification task, the project must evaluate whether selected algorithms can produce useful predictive signals. This is the central technical question of the capstone.

The third question strengthens the business value of the project. A model by itself is not enough; the findings must be communicated clearly. Reporting and visualization help translate predictive results into forms that a manager, analyst, or lending stakeholder can understand and act upon.

Together, these questions are well aligned with the project goal, the selected dataset, and the capstone's expectation that the project combine machine learning, interpretation, and communication of findings.

6. METHODOLOGY AND JUSTIFICATION

The proposed methodology follows a logical sequence from dataset understanding to predictive modeling and reporting. This structure is consistent with the course expectation that Milestone 2 present a tentative approach and preliminary analysis path.

6.1 Exploratory Data Analysis (EDA)

The first stage will be exploratory data analysis. This will include:

- examining target distribution,
- reviewing missingness,
- summarizing important numerical variables,
- visualizing categorical breakdowns,
- checking correlations or associations,
- identifying potential anomalies and outliers.

EDA is necessary because it supports better preprocessing, better feature selection, and stronger interpretation of model outcomes. It also helps ensure that later modeling decisions are based on observed data characteristics rather than assumptions.

6.2 Preprocessing

After EDA, the data will be preprocessed for modeling. This includes:

- missing value treatment,
- categorical encoding,
- scaling where appropriate,
- aggregation of related tables,
- optional feature selection or dimensionality reduction if needed.

This step is justified because credit-risk datasets often contain heterogeneous variables, missing fields, and mixed scales. Clean preprocessing improves both model quality and result reliability. Prior work in credit-risk prediction also emphasizes the importance of preprocessing and feature engineering as core parts of the predictive pipeline.

6.3 Baseline and Comparative Models

The project will begin with an interpretable baseline model such as **logistic regression**. Logistic regression is useful because it offers a clear reference point and is widely recognized in credit scoring. After establishing the baseline, the project will compare it with more flexible models such as:

- **Decision Tree**
- **Random Forest**

- **XGBoost or another boosting-based method**, if feasible in the project environment

This comparison is justified by the literature, which shows that stronger nonlinear models often improve default prediction performance, while simpler models remain valuable for interpretability and benchmarking.

6.4 Model Evaluation

Because credit-risk datasets can be imbalanced, model evaluation will not rely only on accuracy. Instead, the project will use a set of metrics, including:

- **Precision**
- **Recall**
- **F1-score**
- **ROC-AUC**
- possibly **confusion matrix analysis**

This is important because a model can appear accurate while still performing poorly on the minority default class. In a credit-risk setting, recall for risky borrowers and the balance between false positives and false negatives are especially important. The Milestone 1 proposal already identified accuracy, precision, recall, and ROC-AUC as evaluation metrics, and this milestone strengthens that plan by explaining why multiple metrics are necessary.

6.5 Explainability

To answer the first research question and support transparent decision-making, the project will include model explanation techniques. Feature importance from tree-based models and SHAP-based explanations, if feasible, will be used to identify which variables most influence predicted default risk. This is justified by the literature on explainable AI in credit scoring, which stresses that predictive power should be balanced with interpretability and transparency.

6.6 Reporting and Visualization

The final analytical stage will focus on reporting. The results will be translated into clear visual outputs such as:

- class distribution charts,
- missing-value summaries,
- top feature importance charts,
- model comparison tables,
- ROC curves,
- dashboard-style summaries of applicant risk segments.

This stage directly supports the third research question and aligns with the capstone emphasis on communicating findings effectively through visual and written presentation. The overall workflow of the proposed analytical process is summarized in Figure 8.

Figure 8. Overall workflow of the credit risk analysis process

7. ALIGNMENT BETWEEN LITERATURE, DATA, AND METHOD

The proposed approach is appropriate because it connects the literature review, the characteristics of the dataset, and the practical goals of the project.

From the literature, three lessons stand out:

1. credit default prediction is a valid and widely studied classification problem,
2. class imbalance and preprocessing matter, and
3. explainability should accompany performance-focused modeling.

From the dataset, the Home Credit data provide enough variety and scale to support feature analysis, predictive modelling, and reporting. The multi-table structure also creates an opportunity to enrich the applicant-level view with behavioral history.

From the project perspective, the selected methodology is feasible within the capstone scope because it starts with interpretable baseline models, uses standard Python-based tools, and gradually expands toward more advanced models and explanations where feasible. This is consistent with the project design already approved in Milestone 1.

The selection of classification models and preprocessing techniques is informed by prior research, which emphasizes the importance of handling class imbalance, applying interpretable models, and comparing multiple algorithms to achieve optimal predictive performance.

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8. PRELIMINARY CONCLUSION

This milestone has established the methodological foundation for the capstone project. The literature review shows that credit-risk prediction is a mature but still evolving area, especially in relation to class imbalance, predictive performance, and explainability. The Home Credit Default Risk dataset is well suited to this project because it provides structured application and repayment-related information for a realistic classification problem. The proposed methodology addresses both analytical rigor and business usefulness by combining preprocessing, model comparison, explainability, and reporting.

The next phase of the project will implement this approach through coding, exploratory analysis, model training, and preliminary result interpretation.

9. REFERENCES

Use these on your references page in APA style.

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Access to the dataset requires a Kaggle account and acceptance of competition rules.